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Song et al.

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(54) **HIGH SOLDERING STRENGTH TERMINAL FOR SURGE PROTECTION DEVICE**

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H01C 1/144 (2006.01)

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CPC **H01C 1/144** (2013.01); **H01H 37/761** (2013.01); **H01H 2037/762** (2013.01); **H01H 2037/769** (2013.01)

(58) **Field of Classification Search**

CPC H01C 1/144; H01H 37/761; H01H 2037/762; H01H 2037/769; H01H 2037/763; H01H 85/143
See application file for complete search history.

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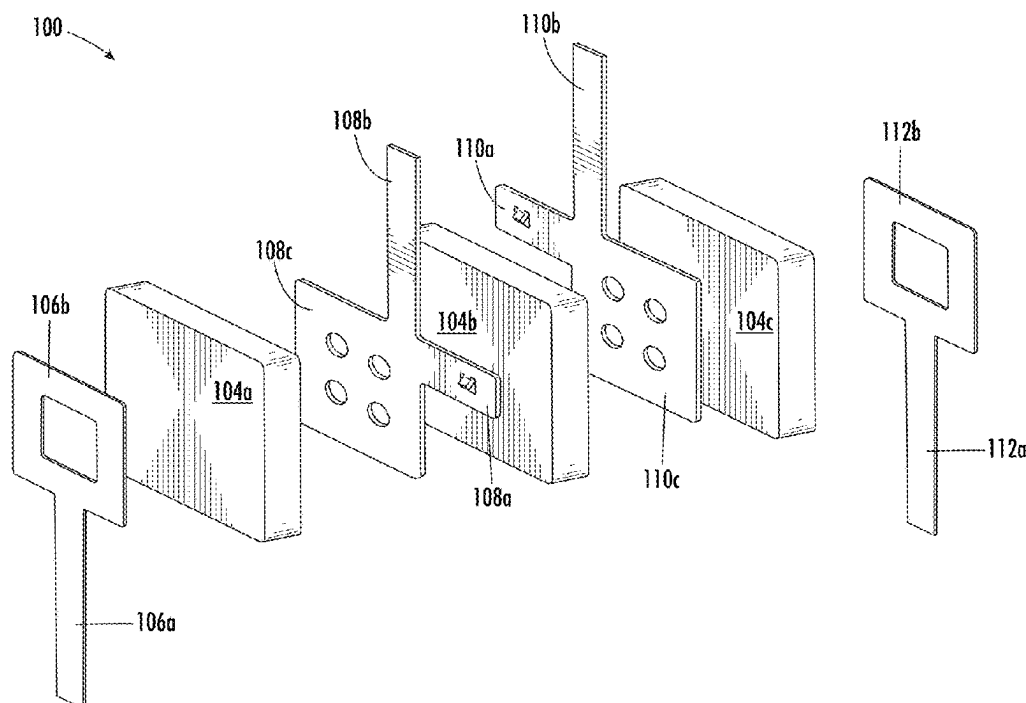
Primary Examiner — Jacob R Crum

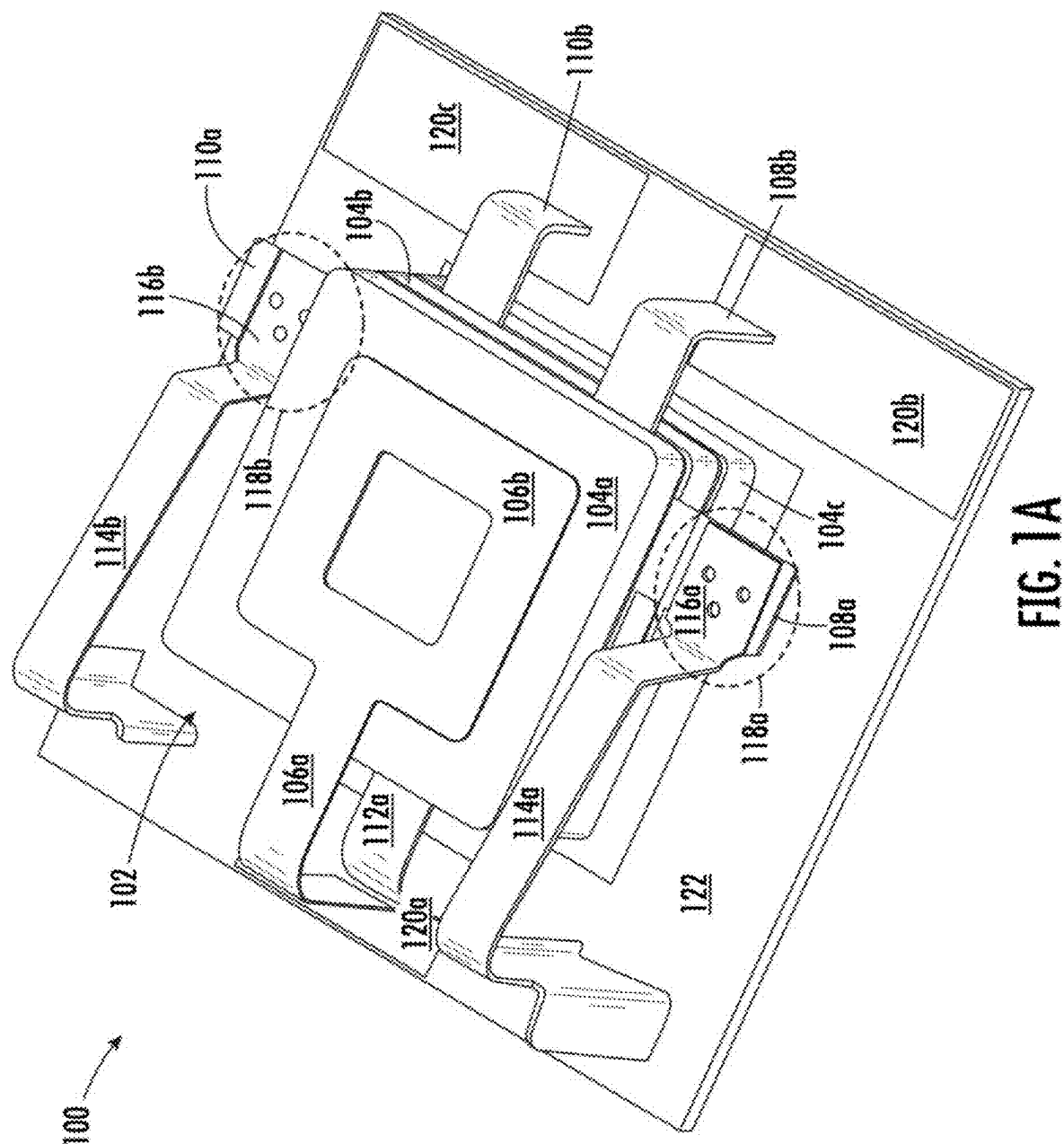
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(57) **ABSTRACT**

A surge protection device includes a metal terminal and a spring. The metal terminal is located between two metal oxide varistors, where the metal terminal extends beyond the metal oxide varistors. The spring has a flat surface that is connected to the metal terminal using a soldering paste. The flat surface has a bend on one end which forms a gap between the flat surface and the metal terminal.

9 Claims, 7 Drawing Sheets





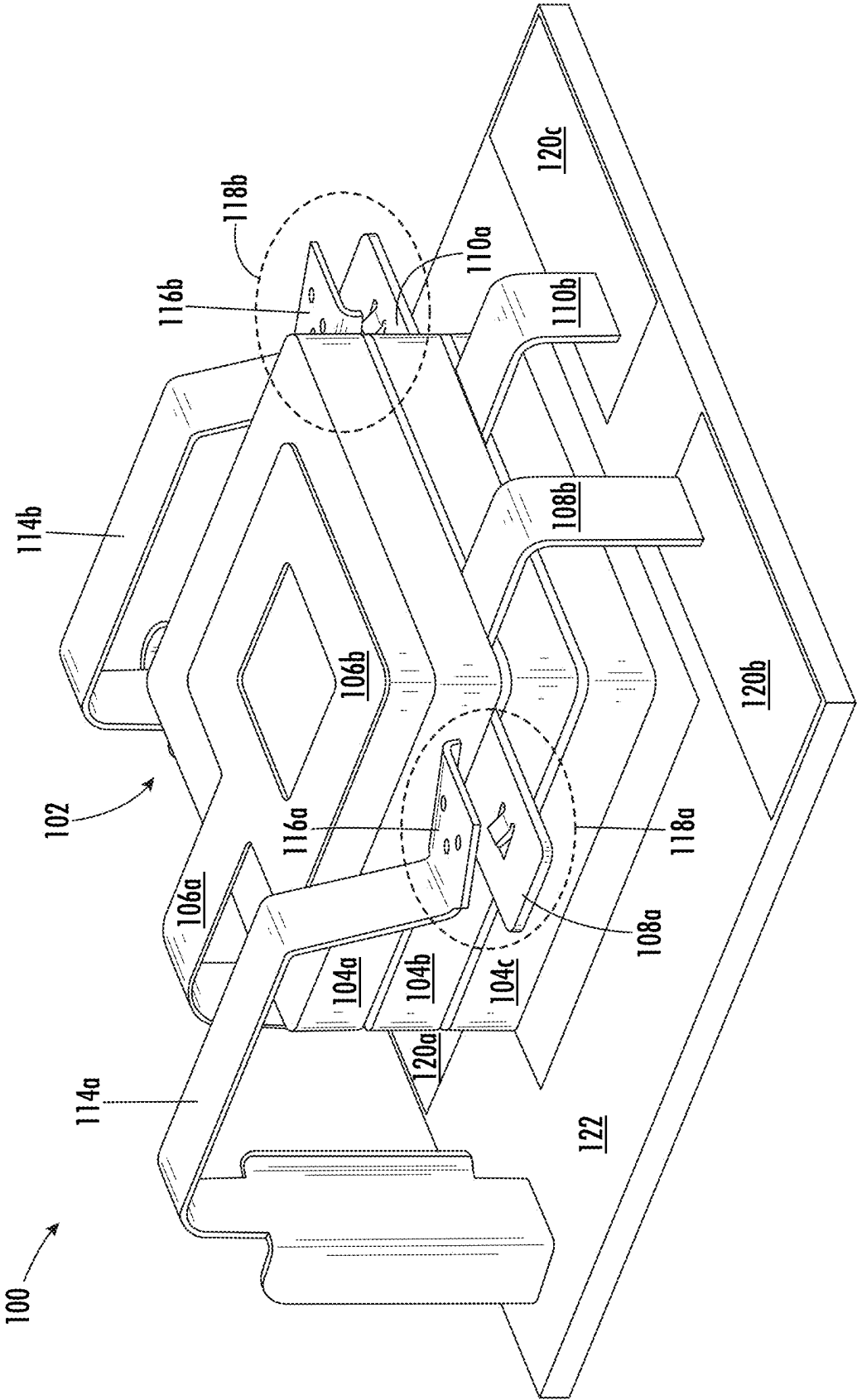


FIG. 1B

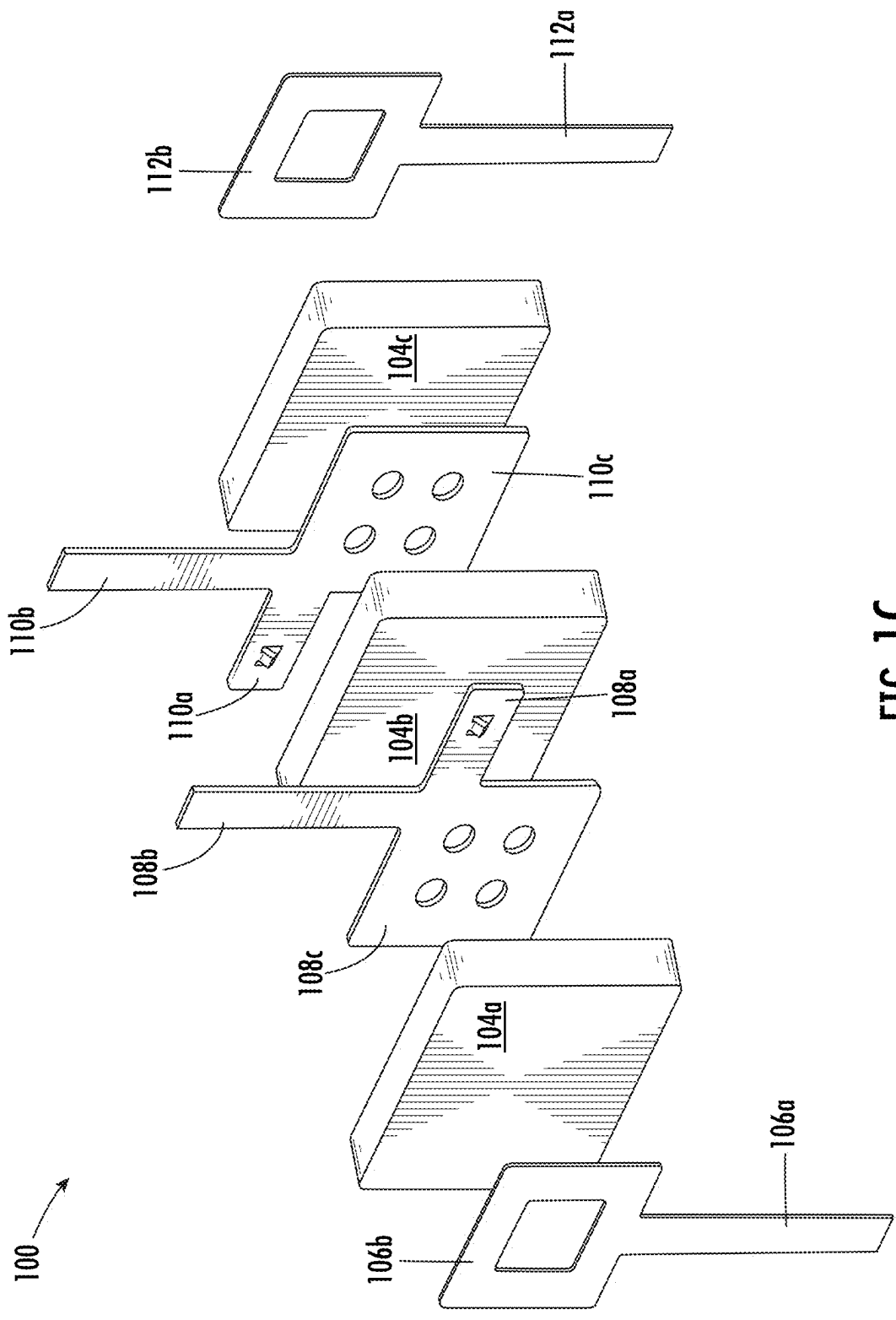


FIG. 1C

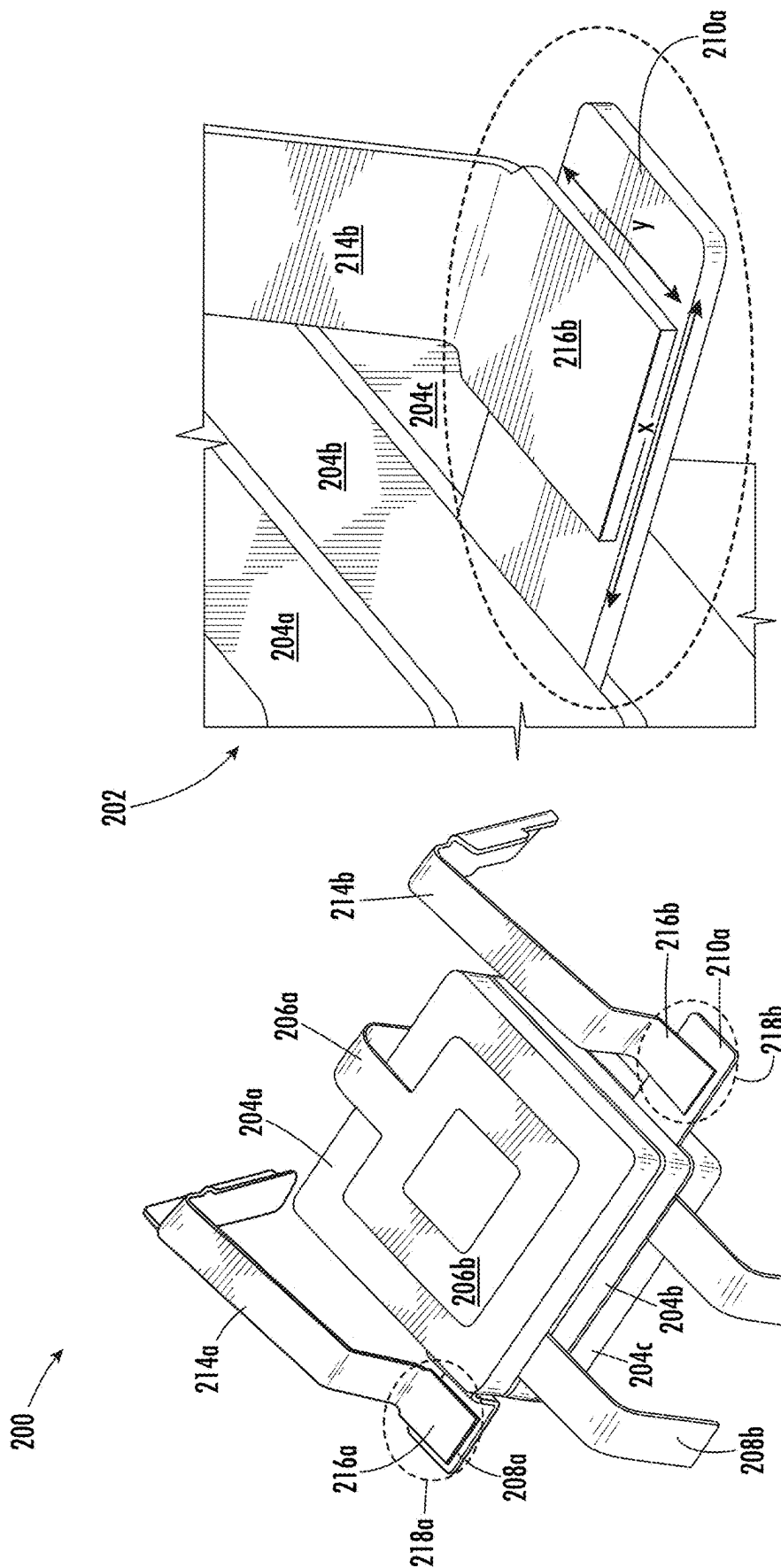
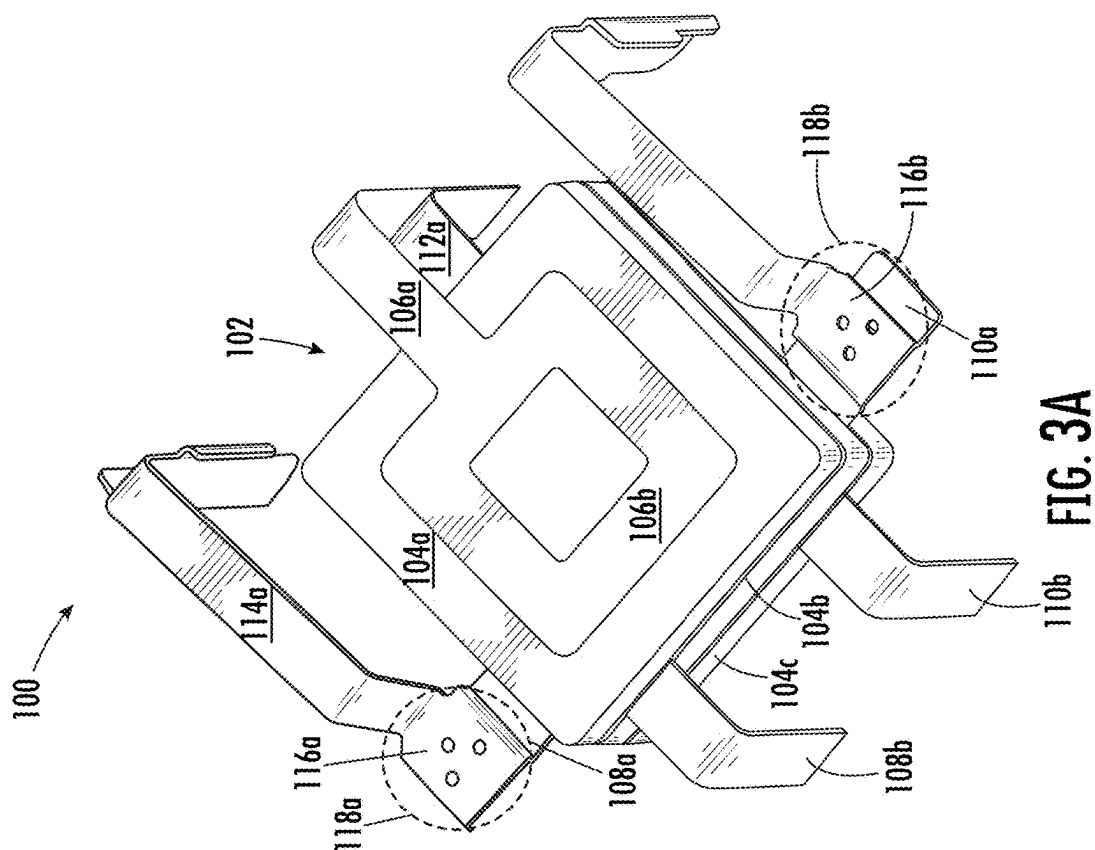
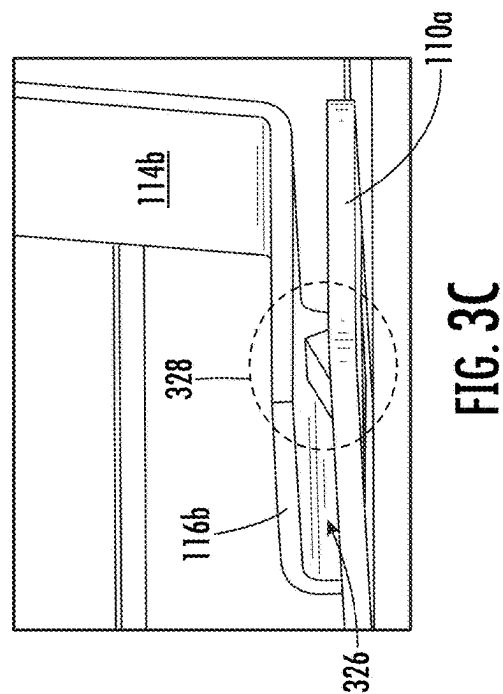
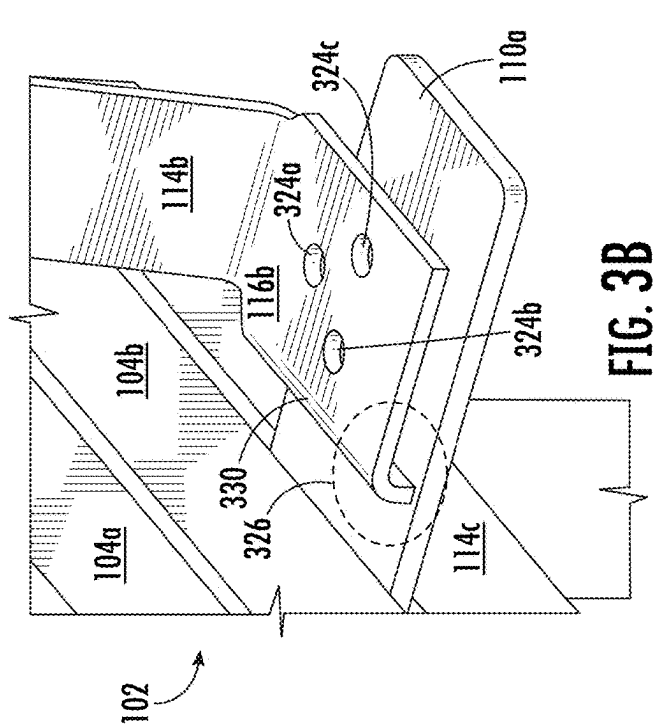


FIG. 2B
(PRIOR ART)

FIG. 2A
(PRIOR ART)



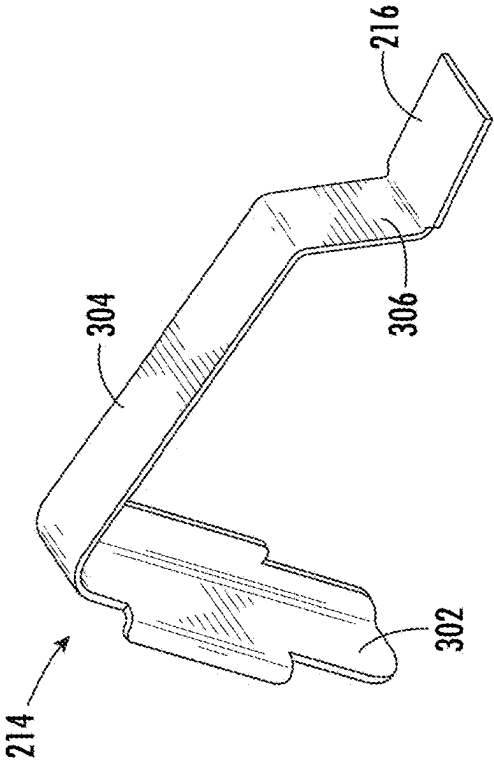


FIG. 4A
(PRIOR ART)

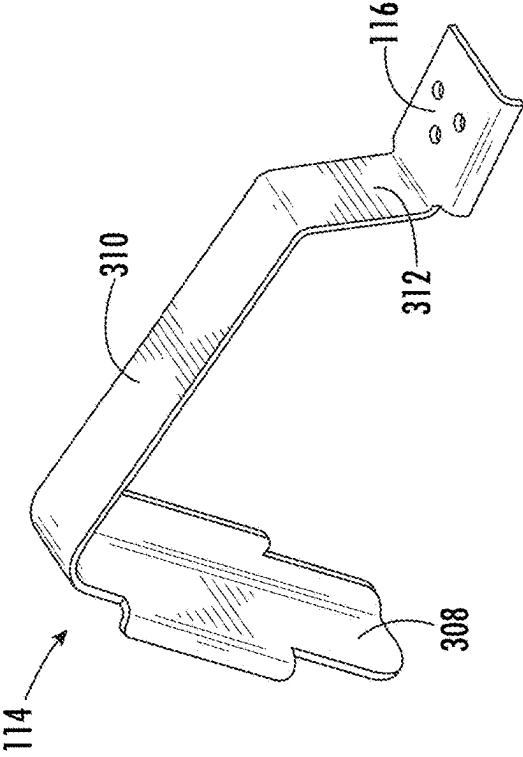


FIG. 4B

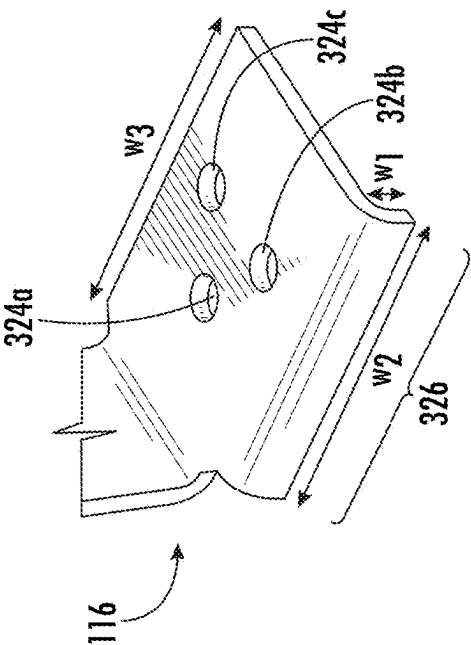


FIG. 4C

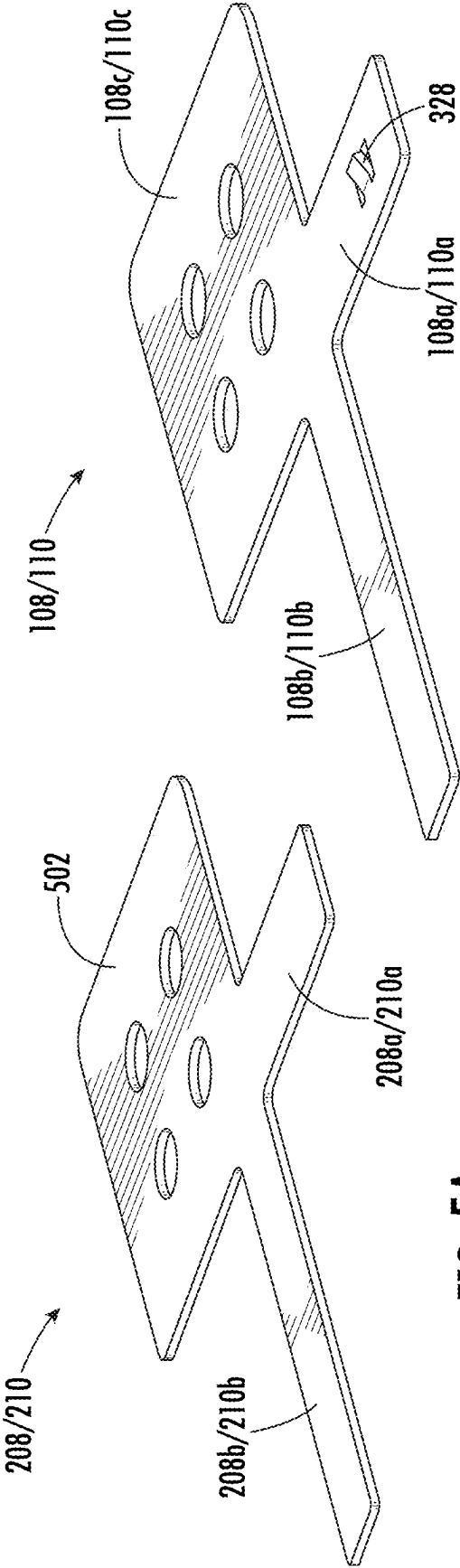


FIG. 5B

FIG. 5A
(PRIOR ART)

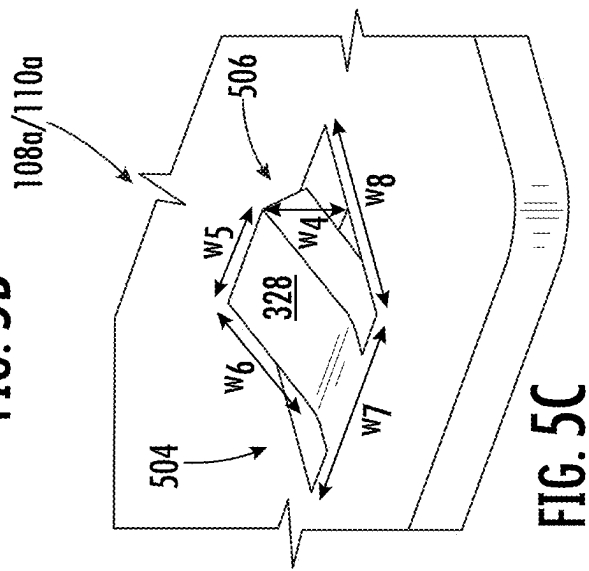


FIG. 5C

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**HIGH SOLDERING STRENGTH TERMINAL
FOR SURGE PROTECTION DEVICE****CROSS REFERENCE TO RELATED
APPLICATION**

This application claims the benefit of priority to, Chinese Patent Application No. 202210108518.1, filed Jan. 28, 2022, entitled “HIGH SOLDERING STRENGTH TERMINAL FOR SURGE PROTECTION DEVICE,” which application is incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

Embodiments of the present disclosure relate to the field of circuit protection and, more particularly, to surge protection devices that employ thermal cutoff.

BACKGROUND

Overvoltage protection devices are used to protect electronic circuits and components from damage due to overvoltage fault conditions. Known variously as surge protection devices (SPDs), lightning surge protection (LSP) devices, and thermal cutoff (TCO) devices, some of these devices employ metal oxide varistors (MOVs). Varistors are voltage dependent, nonlinear devices composed primarily of ZnO with small additions of other metal oxides such as Bismuth, Cobalt, Manganese, and others, resulting in a crystalline microstructure that allows the MOV to dissipate very high levels of transient energy across the entire bulk of the device. MOVs are typically used for the suppression of lightning and other high energy transients found in industrial or AC line applications. Additionally, MOVs are used in DC circuits such as low voltage power supplies and automobile applications.

Some SPDs utilize spring elements soldered to the electrode of the MOV. When an abnormal condition occurs, the solder melts and the spring moves, resulting in an open circuit. In particular, when a voltage that is larger than the nominal or threshold voltage is applied to the device, current flows through the MOV, which generates heat. This causes the linking element, the solder, to melt. Once the link melts, the spring separates from the MOV electrode and an open circuit results, which prevents the MOV from catching fire.

The spring is thus considered an important element in the SPD. A reliable thermal link between the spring and the MOV electrode ensures the formation of an open circuit under the overvoltage condition. The traditional soldering interface between the spring and the MOV electrode may be deficient in terms of the volume of solder paste and the uniformity of application, resulting in unstable soldering strength, which compromises the operation of the SPD.

It is with respect to these and other considerations that the present improvements may be useful.

SUMMARY

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended as an aid in determining the scope of the claimed subject matter.

An exemplary embodiment of a surge protection device in accordance with the present disclosure may include a metal terminal and a spring. The metal terminal is located between

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two metal oxide varistors, where the metal terminal extends beyond the metal oxide varistors. The spring has a flat surface that is connected to the metal terminal using a soldering paste. The flat surface has an opening to allow air bubbles forming in the soldering paste to be released during the connection.

Another exemplary embodiment of a surge protection device in accordance with the present disclosure may include a metal terminal and a spring. The metal terminal is located between two metal oxide varistors, where the metal terminal extends beyond the metal oxide varistors. The spring has a flat surface that is connected to the metal terminal using a soldering paste. The flat surface has a bend on one end which forms a gap between the flat surface and the metal terminal.

An exemplary embodiment of surge protection device in accordance with the present disclosure may include a spring and a metal terminal. The spring has a flat surface. The metal terminal is located between two metal oxide varistors, where the metal terminal extends beyond the metal oxide varistors. The metal terminal is connected to the flat surface using a soldering paste and has a protrusion that forms a gap between the flat surface and the metal terminal.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1C are diagrams illustrating a surge protection device, in accordance with exemplary embodiments;

FIGS. 2A-2B are diagrams illustrating a surge protection device, in accordance with the prior art;

FIGS. 3A-3C are diagrams further illustrating the surge protection device of FIGS. 1A-1C, in accordance with exemplary embodiments;

FIG. 4A is a diagram illustrating the spring used by the surge protection device of FIGS. 2A-2B, in accordance with the prior art;

FIGS. 4B-4C are diagrams illustrating the spring used by the surge protection device of FIGS. 1A-1C, in accordance with exemplary embodiments;

FIG. 5A is a diagram illustrating the metal terminal used by the surge protection device of FIGS. 2A-2B, in accordance with the prior art; and

FIGS. 5B-5C are diagrams illustrating the metal terminal used by the surge protection device of FIGS. 1A-1C, in accordance with exemplary embodiments.

DETAILED DESCRIPTION

A surge protection device with a thermal cutoff spring is disclosed. Within the soldering joints of the surge protection device are three features to facilitate successful placement of soldering paste between two flat surfaces: one or more openings, a bend, and a protrusion. The one or more openings and the bend are on the flat surface of a spring used for thermal cutoff. The protrusion is part of a metal terminal to which the flat surface is connected. The openings provide an exit for air bubbles within the soldering paste that may occur. The protrusions and the bend provide gaps between the two flat surfaces, which facilitate both control of the amount of soldering paste and targeted positioning of the paste.

For the sake of convenience and clarity, terms such as “top”, “bottom”, “upper”, “lower”, “vertical”, “horizontal”, “lateral”, “transverse”, “radial”, “inner”, “outer”, “left”, and “right” may be used herein to describe the relative placement and orientation of the features and components, each with respect to the geometry and orientation of other fea-

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tures and components appearing in the perspective, exploded perspective, and cross-sectional views provided herein. Said terminology is not intended to be limiting and includes the words specifically mentioned, derivatives therein, and words of similar import.

FIGS. 1A-1C are representative drawings of a surge protection device (SPD) 100 for providing overvoltage protection, according to exemplary embodiments. FIG. 1A is a perspective view of the SPD 100 in a first state; FIG. 1B is a perspective view of the SPD in a second state; and FIG. 1C is an exploded perspective view of the SPD. The SPD 100 consists of metal oxide varistors (MOVs) arranged as an MOV stack 102, with metal terminals disposed above, below, and between the MOVs. In exemplary embodiments, the SPD 100 consists of three MOVs 104a, 104b, and 104c (collectively, "MOVs 104") although the SPD may have more or fewer MOVs than are shown.

Above, below, and between the MOVs 104 are metal terminals, with the number of metal terminals being dependent on the number of MOVs. The exploded view of FIG. 1C provides a full view of the metal terminals: a first metal terminal consists of a long terminal 106a and a body 106b (collectively, "metal terminal 106"); a second metal terminal consists of a short terminal 108a, a long terminal 108b, and a body 108c (collectively, "metal terminal 108"); a third metal terminal consists of a short terminal 110a, a long terminal 110b, and a body 110c (collectively, "metal terminal 110"); and a fourth metal terminal consists of a long terminal 112a and a body 112b (collectively, "metal terminal 112"). Although distinct portions are called out in the figures, each metal terminal 106, 108, 110, and 112 is formed as a unitary structure of electrically conductive material, such as copper, in exemplary embodiments.

With reference also to FIGS. 1A-1B, metal terminal 106 is disposed atop the MOV stack 102, specifically, adjacent the MOV 104a; metal terminal 108 is disposed between MOV 104a and MOV 104b, with the body 108c being sandwiched between the MOVs and the short terminal 108a and the long terminal 108b extending beyond the structure of the MOVs; metal terminal 110 is disposed between MOV 104b and MOV 104c, with the body 110c being sandwiched between the MOVs and the short terminal 110a and the long terminal 110b extending beyond the structure of the MOVs; and metal terminal 112 is disposed at the bottom of the MOV stack 102, specifically, adjacent the MOV 104c. The long terminals 106a, 108b, 110b, and 112a are to be bent for connection to a PCB 122. Long terminals 106a and 112a are bent and connected to a pad 120a; long terminal 108b is bent and connected to a pad 120b and long terminal 110b is bent and connected to a pad 120c (collectively, "pads 120"). In exemplary embodiments, the long terminals 106a, 108b, 110b, and 112a are soldered to respective pads 120.

In exemplary embodiments, the SPD 100 also includes a pair of springs 114a and 114b (collectively, "springs 114"). Each spring 114 features a flat surface for coupling (soldering) to metal terminals: flat surface 116a of spring 114a is soldered to metal terminal 108a and flat surface 116b of spring 114b is soldered to metal terminal 110a (collectively, "flat surfaces 116"). The springs 114 provide a mechanism by which the SPD 100 is uncoupled (released) from connection to a device, such as a power supply in response to an overvoltage condition.

In FIGS. 1A and 1B, a circled soldering joint 118a includes the flat surface 116a of the spring 114a and the metal terminal 108a; similarly, a circled soldering joint 118b includes the flat surface 116b and the metal terminal 110a (collectively, "soldering joints 118"). The flat surface 116a

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of spring 114a is affixed to the metal terminal 108a and the flat surface 116b of spring 114b is affixed to the metal terminal 110a using a soldering paste. The flat surface 116b is disposed over and close to the metal terminal 110a.

In exemplary embodiments, the soldering paste is selected to have a particular melting point that is consistent with the rating of the SPD 100. For example, the soldering paste may be low melting point solder paste consisting of 42% tin (Sn) and 58% bismuth (Bi), which has a melting point of 138° C. Until an overvoltage condition for which the SPD 100 is designed occurs, current flows through the SPD (normal operation). FIG. 1A shows the SPD 100 during normal operation, with spring 114a soldered to the metal terminal 108a and spring 114b soldered to the metal terminal 110a.

To provide surge protection, the SPD 100 does the following: An overvoltage condition occurs, causing the MOV stack 102 to overheat. Heat is transferred to the metal terminals 106, 108, 110, and 112, with more substantial heat being transferred to the metal terminals that are in between MOVs 204, namely, the metal terminals 108 and 110. Respective soldering joints 118a and 118b melt, causing the flat surface 116a of spring 114a to move away from the metal terminal 108a and the flat surface 116b of spring 114b to move away from the metal terminal 110a. FIG. 1B shows the SPD 100 during the overvoltage event, with spring 114a decoupled from the metal terminal 108a and spring 114b decoupled from the metal terminal 110a. The distance between the flat portions 116 and respective metal terminals 108a and 110a may form a gap of between 3 and 8 mm, for example. This results in an open circuit between the SPD 100 and the circuit to which the SPD is connected, such as a power supply. The MOV stack 102 is thus able to cool down, with the sequence of operations preventing the SPD 100 from burning.

The springs 114 of the SPD 100 are designed to become uncoupled from respective metal terminals upon occurrence of an overvoltage event for which the SPD is rated. Once the overvoltage event occurs, the flat surface 116a of the spring 114a becomes uncoupled from the metal terminal 108a and the flat surface 116b of the spring 114b becomes uncoupled from the metal terminal 110a. In this way, an open circuit is formed.

SPD devices are thus successful by having a reliable thermal link between the flat portion of the spring and the metal electrode to which they are affixed to form open circuit under the overvoltage condition. This depends on having the correct soldering paste for the overvoltage characteristics of the SPD device, as well as the proper placement and quantity of soldering paste. In exemplary embodiments, the SPD 100 includes features in the soldering joints 118 that improve the quality of soldering between the flat surfaces and the metal terminals. To understand and appreciate the improved features, a discussion of an SPDs according to the prior art is appropriate.

FIGS. 2A-2B are representative drawings of a surge protection device (SPD) 200, according to the prior art. FIG. 2A is a perspective view of the SPD 200 and FIG. 2B is a closeup perspective view of a component of the SPD. The SPD 200 consists of an MOV stack 202 with metal terminals disposed above, below, and between the MOVs. The SPD 200 consists of three MOVs 204a, 204b, and 204c (collectively, "MOVs 204") although the SPD may have more or fewer MOVs than shown.

Above, below, and between the MOVs 204 are metal terminals, with the number of metal terminals being dependent on the number of MOVs. Although there is no exploded view of the SPD 200, the configuration of metal terminals

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and MOVs is similar to that of the exemplary SPD 100, with the visible portions of the metal terminals being called out herein. A first metal terminal consists of a long terminal 206a and a body 206b (collectively, “metal terminal 206”); a second metal terminal consists of a short terminal 208a and a long terminal 208b (collectively, “metal terminal 208”); a third metal terminal consists of a short terminal 210a and a long terminal 210b (collectively, “metal terminal 210”).

Metal terminal 206 is disposed atop the MOV stack 202, specifically, adjacent the MOV 204a; metal terminal 208 is disposed between MOV 204a and MOV 204b; metal terminal 210 is disposed between MOV 204b and MOV 204c; a fourth terminal (not shown) is disposed at the bottom of the MOV stack 202, specifically, adjacent the MOV 204c. The long terminals 206a, 208b, and 210b are to be bent for connection to a PCB (not shown).

As with the exemplary SPD 100, the prior art SPD 200 includes a pair of springs 214a and 214b (collectively, “springs 214”). Each spring 214 features a flat surface for soldering to metal terminals: flat surface 216a of spring 214a is soldered to metal terminal 208a and flat surface 216b of spring 214b is soldered to metal terminal 210a (collectively, “flat surfaces 216”).

In FIG. 2A, a circled soldering joint 218a includes the flat surface 216a of the spring 214a and the metal terminal 208a; similarly, a circled soldering joint 218b includes the flat surface 216b and the metal terminal 210a (collectively, “soldering joints 218”). FIG. 2B provides a closeup view of the soldering joint 218b. The flat surface 216a of spring 214a is affixed to the metal terminal 208a and the flat surface 216b of spring 214b is affixed to the metal terminal 210a using a soldering paste. The flat surface 216b is disposed over and close to the metal terminal 210a, with the soldering paste being selected to have a particular melting point that is consistent with the rating of the SPD 200.

As with the SPD 100 (FIGS. 1A-C), the SPD 200 may be disposed on a printed circuit board (not shown) and connected to an electronic device, such as a power supply. Until an overvoltage condition for which the SPD 200 is designed occurs, current flows through the SPD as normal. But during an overvoltage event, the MOV stack 202 overheats, heat is transferred to the metal terminals, causing the soldering joints 218 to melt, with the flat portions 216 of each spring 214 to move away from respective metal terminals 208a and 210a, resulting in an open circuit between the SPD 200 and the device to which it is connected, such as a power supply. The MOV stack 202 is able to cool down and a fire hazard or destruction of either the SPD 200, the power supply, or both, are avoided.

SPD devices are thus successful by having a reliable thermal link between the flat portion of the spring and the metal electrode to which they are affixed to form an open circuit during the overvoltage event. The reliable thermal link depends on having the correct soldering paste for the overvoltage characteristics of the SPD device. Further, the traditional soldering interface for the SPD 200 is for a first flat surface to directly touch a second flat surface. FIG. 2B is instructive in illustrating the drawbacks of the prior art SPD 200.

First, the volume of solder paste cannot be precisely controlled and easily be pushed away during application on the prior art SPD 200. The solder paste is to be applied to the metal terminal 210a, then the flat surface 216b of the spring 214b is positioned over the soldering paste. The soldering paste is thus sandwiched between the metal terminal 210a and the flat surface 216b. Depending on the precise application of the flat surface 216b on the soldering paste, the

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pressure applied to the metal terminal 210a, and other factors, the soldering paste may ooze out between the two surfaces, resulting in an inconsistent amount of soldering paste in between the two flat surfaces. By not having a consistent amount of soldering paste at the soldering interface, an unstable soldering strength may result.

Second, the application of the soldering paste may result the formation of air bubbles between the two flat surfaces, the metal terminal 210a and the flat surface 216b. It is difficult to determine whether there are air bubbles during application because the flat surface 216b of the spring 214b is not transparent. Further, even when the presence of the air bubbles is known, it is difficult to eliminate the air bubbles from forming between the two flat surfaces. The presence of air bubbles may depend to some extent on the chemical makeup of the soldering paste. Since the soldering paste is formulated based on a desired melting point of the soldering paste and thus voltage rating of the SPD, the air bubbles present an additional condition that negatively affects the soldering strength and reliability.

Third, there are two flat surfaces, the metal terminal 210a, which is stationary, and the flat surface 216b of the spring 214b, which, until the soldering paste is applied, is not stationary. Further, the shape of the flat surfaces is not identical: the rectangular shape of the metal terminal 210a has a larger area than the rectangular shape of the flat surface 216b. Thus, during application of the soldering paste, the flat surface 216b may move closer to the MOVs 204b and 204c or farther away from the MOVs, as shown by arrow, x. Or the flat surface 216b may move in parallel to the MOVs 204b and 204c but move in the directions shown by arrow, y (orthogonal to x). Even still, the flat surface 216b may move so that it is not orthogonal to the metal terminal 210a. Because the flat surfaces of the metal terminal 210a and the flat surface 216b are able to easily slide relative to one other and are not fixed, the resulting soldering strength is unpredictable. Any one of the above problems may produce defective soldering, and thus deficient operation of the prior art SPD 200.

FIGS. 3A-3C are representative drawings of the SPD 100 already introduced in FIGS. 1A-1C, according to exemplary embodiments. FIG. 3A is a perspective view of the SPD 100 and FIGS. 3B-3C are closeup perspective views of the novel features of the SPD. In addition to the aforementioned features, the SPD 100 includes novel elements, located within the soldering joints 118: openings 324a-c (collectively, “openings 324”), a bend 326, and a protrusion 328. In exemplary embodiments, the novel elements solve the problems found with the application of soldering paste in the SPD 200 described above.

In exemplary embodiments, the openings 324 are located in the flat surfaces 116 of the springs 114. Although three openings 324a, 324b, and 324c are shown, the flat surfaces 116 may have more or fewer openings. Further, the openings 324 do not have to be circular, they may any shape or size depending on what is needed. A soldering paste made of a thicker composition of materials may warrant having larger openings, for example. In exemplary embodiments, as the soldering paste is applied between the flat surfaces 116 and the metal terminals 108a and 110a, the openings 324 provide a pathway in which air bubbles formed in the soldering paste are able to escape.

Using the spring 114a and the metal terminal 108a as an example, the soldering paste is applied to the metal terminal 108a, then the flat surface 116a of the spring 114a is disposed over and pressed toward the metal terminal. If air bubbles form within the soldering paste, the openings 324

provide an escape route for the air bubbles to move upward during the downward press of the flat surface **116a** onto the metal terminal **108a**. The openings **324** thus help to void air bubbles that form inside the soldering joints **118**.

In exemplary embodiments, the bend **326** is a modification of the flat surface **116** of the spring **114**. Both springs **114** include bends **326**, though one is featured in the detailed drawings of FIGS. 3B and 3C. Bend **326** extends along the length of one side **330** of the flat surface **116b**, in exemplary embodiments. Alternatively, the bend **326** may extend along other sides of the flat surface **116b**. The bend **326** is shown and described in more detail in FIGS. 4B and 4C, below. In exemplary embodiments, the bend **326** provides a gap of a predetermined size between the metal terminal **110a** and the flat surface **116b** of the spring **114b**. Further, in exemplary embodiments, the bend **326** helps to control the amount (a quantity) of soldering paste used to secure the two flat surfaces together. The bend **326** provides a barrier to prevent the soldering material from oozing out from between the flat surfaces. The bends **326** thus help to maintain a specific gap between the metal terminal and flat surface of the spring to ensure there is enough solder paste to provide a desired soldering strength, which helps to precisely control the usage of soldering paste.

While the openings **324** and the bend are part of the springs **114**, the protrusions **328** are part of the metal terminals **108a** and **110a**. As illustrated in FIG. 3C, the protrusion **328** is a raised structure of the metal terminal **110a**. In exemplary embodiments, the protrusion **328** provides a gap between the two flat structures. The protrusion **328** is shown and described in more detail in FIGS. 5B and 5C, below. The protrusion **328** thus provides an adjustable distance for soldering paste feeding, which helps to precisely control the remaining soldering paste amount in the soldering interface. Recall from FIG. 2B how movement of the flat surface **216b** of the prior art SPD **200** is difficult to control during the soldering paste operation. In exemplary embodiments, the protrusion **328** acts as a stopper to limit movement of the flat surface **116** of the spring **114**, which helps to fix the soldering field.

FIGS. 4A-4C are representative drawings comparing the novel spring **114** (SPD **100**) with the prior art spring **214** (SPD **200**), according to exemplary embodiments. FIG. 4A is the spring **214** of the prior art SPD **200** (FIGS. 2A-2B); FIG. 4B is the spring **114** of the novel SPD **100** (FIGS. 1A-1C and 3A-3C); and FIG. 4C is a closeup view of the flat surface of the novel spring.

The prior art spring **214** is illustrated in FIG. 4A and the novel spring **114** is illustrated in FIG. 4B. The springs **114** and **214** are made of a rigid material, such as beryllium copper, steel, aluminum, or alloys of steel and aluminum having several pieces that are bent to enable a spring-like response. Prior art spring **214** includes first portion **302**, second portion **304**, third portion **306**, and the already introduced flat surface **216**.

The first portion **302** touches and rests on the surface to which the SPD **200** is mounted, such as a printed circuit board. The second portion **304** is somewhat orthogonal to the first portion **302** and is connected between the first portion and the third portion **306**. The third portion **306** is bent relative to the second portion **304** at an obtuse angle (e.g., greater than 90 degrees). The flat surface **216**, as already shown and described, is designed to be soldered to a metal electrode of the SPD **200**, and is disposed at an obtuse angle relative to the third portion **306**. The four portions are arranged so that the flat surface **216** will move

upward in response to an overvoltage event, thus disengaging the flat surface from the metal electrode of the SPD **200**.

The novel spring **114** in FIG. 4B is similarly made up of four portions, a first portion **308**, which is to sit on the PCB (see, e.g., FIGS. 1A and 1B), a second portion **310**, a third portion **312**, and the already introduced flat surface **116**. The angular arrangement of the novel spring **114** is similar to that of the prior art spring **214**. In contrast to the prior art spring **214**, the novel spring **114** includes, on the flat surface **116**, the openings **324** and the bend **326** shown also in FIG. 3B.

In exemplary embodiments, the openings **324** help to mitigate the occurrence of air bubbles in the soldering paste that may be disposed between the metal terminal of the SPD **100** and the flat surface **116** while the bend **326** helps to prevent soldering paste oozing from between the flat surface **116** and the metal terminal. Further, as illustrated in FIG. 4C, the bend **326** provides a gap of distance w_3 between the metal terminal and the flat surface **116**. In exemplary embodiments, the gap facilitates controlling the amount of soldering paste that is used to secure the flat surface **116** to the metal terminal of the SPD **100**.

The bend **326** extends along one side of the flat surface **116** and has a width, w_2 while the flat surface has a width, w_3 . In exemplary embodiments, the two widths are the same, that is, $w_2 = w_3$. The bend **326** may alternatively be on another side of the flat surface **116**, whether the side orthogonal or opposite to its current location. Or the spring **114** may have multiple bends disposed on more than one side of the flat surface **116**.

FIGS. 5A-5C are representative drawings comparing the novel metal terminals **108** and **110** (SPD **100**) with the prior art metal terminal **208** and **210** (SPD **200**), according to exemplary embodiments. FIG. 5A is the metal terminal **208/210** of the prior art SPD **200** (FIGS. 2A-2B); FIG. 5B is the metal terminal **108/110** of the novel SPD **100** (FIGS. 1A-1C and 3A-3C); and FIG. 5C is a closeup view of the protrusion **328** of the novel metal terminal.

The prior art metal terminal, which may be either metal terminal **208** or metal terminal **210** (FIGS. 2A-2B), is illustrated in FIG. 5A; the novel metal terminal, which may be either metal terminal **108** or metal terminal **110**, is illustrated in FIG. 5B. The metal terminals **108**, **110**, **208**, and **210** are made of an electrically conductive material, such as copper or copper alloy. The prior art metal terminals **208/210** are made up of the short terminal **208a/210a**, the long terminal **208b/210b** and a body **502**. The short terminal **208a/210a** and the long terminal **208b/210b** stick out from the MOV stack, with the long terminal being bent while the body **502** is disposed between two MOVs (see, e.g., FIG. 2A).

The novel metal terminal is similarly made up of an electrically conductive material, with a short terminal **108a/110a**, a long terminal **108b/110b** and a body **108c/110c**. The short terminal **108a/110a** and the long terminal **108b/110b** stick out from the MOV stack, with the long terminal being bent while the body **108c/110c** is disposed between two MOVs (see, e.g., FIGS. 1A-1B). In contrast to the prior art metal terminals **208/210**, the novel metal terminals **108/110** feature the protrusion **328** disposed on the short terminal **108a/110a**.

In exemplary embodiments, the protrusion **328** provides a gap, w_4 , between the metal terminal **108a/110a** and the flat surface of the spring. Like the bend **326** of the novel spring **114**, the protrusion **328** provides space for a controlled amount of soldering paste to be applied between the flat surface **116** of the spring **114** and the metal terminal **108a/**

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110a. In exemplary embodiments, the gap of the bend **326**, w_1 , is equal to the gap of the protrusion **328**, w_4 , $w_1 = w_4$.

As illustrated in FIG. 5C, the protrusion **328** is created by cutting a rectangular shape **504** having dimensions $w_7 \times w_8$ into the metal terminal **108a/110a** and, from the rectangular shape **504**, cutting another, smaller rectangular shape **506** having dimensions $w_5 \times w_6$, with $w_5 < w_7$ and $w_6 < w_8$. Next, the smaller rectangular shape **506** having dimensions $w_5 \times w_6$ is bent upward by a distance, w_4 to form the protrusion **328**. Because the metal terminal **108a/110a** is a soft metal (copper or copper alloy), the bending of the smaller rectangular shape **506** with dimensions $w_5 \times w_6$ is possible.

In exemplary embodiments, the novel SPD **100** which provides a high soldering strength thermal cutoff terminal can be extended to all flat surfaces of the SPD. The use of the openings **324** and the bend **326** in the spring as well as the protrusion **328** in the metal terminal can be extended to other flat surfaces being connected using soldering paste, not just the springs and the metal terminals. With ease of assembly and low cost, the principles described herein can be applied to a wide variety of metal soldering operations. The result is high soldering performance to enhance soldering strength and product reliability.

As used herein, an element or step recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to “one embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

While the present disclosure makes reference to certain embodiments, numerous modifications, alterations and changes to the described embodiments are possible without departing from the sphere and scope of the present disclosure, as defined in the appended claim(s). Accordingly, it is intended that the present disclosure not be limited to the described embodiments, but that it has the full scope defined by the language of the following claims, and equivalents thereof.

The invention claimed is:

1. A surge protection device comprising:
 - a metal terminal comprising:
 - a body disposed between two metal oxide varistors in direct contact with surfaces of the metal oxide varistors;

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- a short terminal extending beyond outermost peripheries of the metal oxide varistors in a first direction parallel to the surfaces of the metal oxide varistors; and

- a long terminal extending beyond outermost peripheries of the metal oxide varistors in a second direction parallel to the surfaces of the metal oxide varistors and perpendicular to the first direction; and

- a spring comprising a flat surface, the flat surface to be coupled to the metal terminal using a soldering paste, the flat surface further comprising an opening to allow air bubbles forming in the soldering paste to be released during the coupling.

2. The surge protection device of claim 1, the flat surface further comprising a bend disposed on one end of the flat surface, the bend to form a gap between the flat surface and the metal terminal.

3. The surge protection device of claim 2, wherein the gap controls a quantity of soldering paste applied to the metal terminal.

4. The surge protection device of claim 3, the metal terminal further comprising a protrusion, the protrusion to form a second gap between the flat surface and the metal terminal.

5. The surge protection device of claim 4, wherein the second gap controls the quantity of soldering paste applied to the metal terminal.

6. The surge protection device of claim 4, wherein the gap has a first width and the second gap has a second width, the first width being approximately the same as the second width.

7. The surge protection device of claim 1, the metal terminal further comprising a protrusion, the protrusion to form a gap between the flat surface and the metal terminal.

8. The surge protection device of claim 7, wherein the protrusion is a first rectangular shape cut into a second rectangular shape of the metal terminal, wherein the second rectangular shape is larger than the first rectangular shape.

9. The surge protection device of claim 1, wherein the flat surface becomes uncoupled from the metal terminal in response to an overvoltage event for which the surge protection device is rated.

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